AR
2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Aurora Innovation

NASDAQ : AUR · Young-Company DCF (Damodaran)

16 May 2026 · the qualitative

$7.74

$13.2B mkt cap · 1.71B sh · $1.28B cash, $650M net debt

THE CENTRAL QUESTION

Will Aurora capture meaningful share of a $160B US autonomous trucking TAM by 2036 — or is today's $13B market cap pricing certainty about driverless commercial economics that hasn't been proven?

Aurora trades at ~$13.2B against $11M of Q1 2026 revenue and $640M annualized cash burn. The $1.28B liquidity position needs replenishment by mid-2027 absent commercial conversion. Driverless commercial launch slipped from 2024 to now 2027; the Driver runs supervised highway pilots with PACCAR, Volvo, FedEx. Young-company DCF below: TAM share, terminal margin, P(failure). Three scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

$6.65 expected fair value today

-14.0% vs spot $7.74

FORWARD COMPOUNDED VALUE

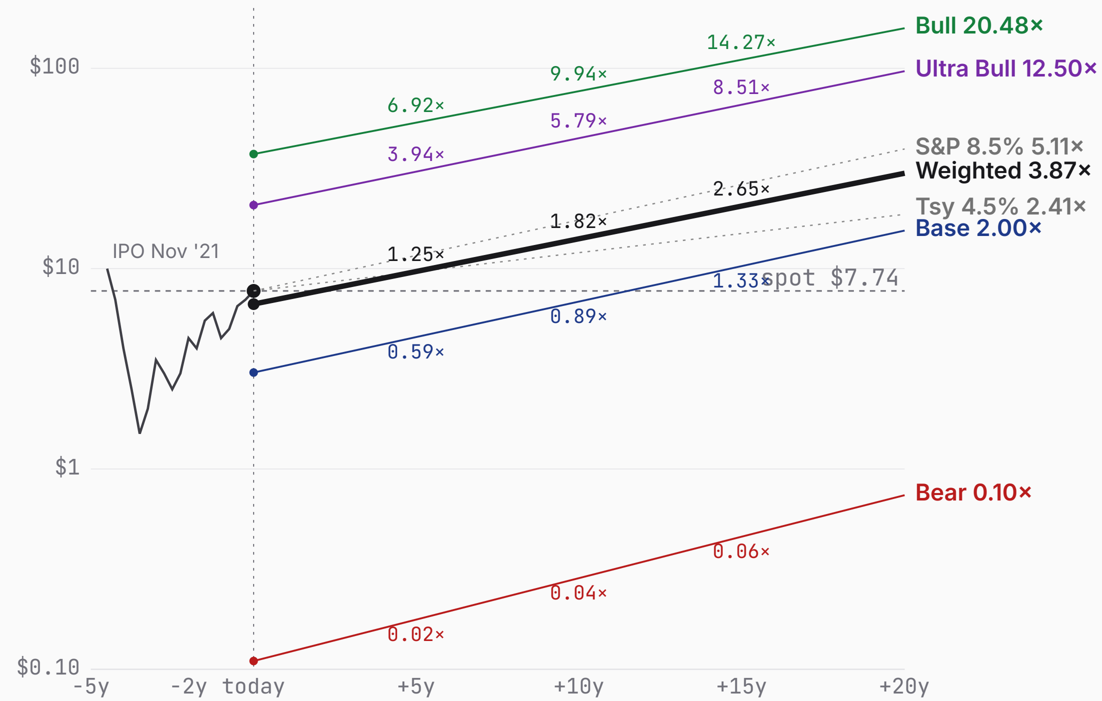
+5y	+10y	+15y	+20y
$9.68	$14.10	$20.54	$29.93
1.25× spot	1.82× spot	2.65× spot	3.87× spot

WEIGHTING RATIONALE

Bear 35% Autonomous trucking unproven at scale; Waymo Via shutdown signal still recent.
Base 50% Modal outcome: Aurora is one of 2-3 winners, 7-8% TAM share, tech-auto blend margins.
Bull 12% Compound conditional chain; software-take-rate platform thesis requires multiple alignments.
Ultra Bull 3% Tail of tails: dominant US autonomous + early software-take-rate + multiple expansion.

Bull (12%) + Ultra Bull (3%) together contribute $5.10 — 77% of the $6.65 expected value despite 15% combined probability. Today's spot ($7.74) sits 16% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



BEAR	BASE	BULL	ULTRA BULL
$0.11	$3.03	$37.32	$20.76
-98.6% vs spot	-60.9% vs spot	+382.2% vs spot	+168.2% vs spot
TAM 1.5% P(fail) 45% S2C 1.2 Prob 35%	TAM 7.5% P(fail) 25% S2C 1.5 Prob 50%	TAM 22.5% P(fail) 12% S2C 1.8 Prob 12%	TAM 6.6% P(fail) 3% S2C 2.5 Prob 3%
Tech doesn't scale, capital exhausts.	Commercial scale works at limited share.	Dominant US platform, software-like margins.	Dominant US; software-take-rate.

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Aurora Innovation scenario narratives

PROBABILITY WEIGHTS

Bear 35% Base 50% Bull 12% Ultra Bull 3% · Weighted expected $6.65 (-14.0% vs spot)

BEAR $0.11

-98.6% vs spot

Probability 35%

Tech doesn't scale, capital exhausts.

WHY 35%

Bearish thesis is compound: tech doesn't scale from supervised highway pilots to driverless at necessary safety thresholds; capital exhausts before commercial validation; dilution at falling prices accelerates as capital needs persist; competitive losses to Kodiak (cargo-focused, well-funded) and Waabi (smaller but capital-efficient). Within-scenario 45% p_fail captures Aurora-specific failure; the 35% scenario weight reflects category-level risk — autonomous trucking economics may not scale for Western operators. Waymo Via's 2023 shutdown was a credible signal of category difficulty. The 35% weight is slightly above JOBY's 30% bear because AUR is earlier-stage (pre-commercial), more capital-dependent, and faces more direct competition.

WHAT HAPPENS

The pessimistic case is brutal. Highway autonomy is harder than demos suggest — five years of 'next year' promises (launch was 2025, then 2026, now 2027), and the long tail of edge cases (night construction, weather, urban surface streets) proves more durable than current systems handle. Commercial routes stay limited to highway-only, daytime, sunbelt-only operation.

Cash burn doesn't moderate. Q1 2026's $640M annualized outflow keeps rising. The $1.28B liquidity needs replenishment by mid-2027. Successive raises come at lower prices — cumulative ~74% dilution by FY36, issuance averaging below $3, destroying per-share value even when the business survives.

BASE $3.03

-60.9% vs spot

Probability 50%

Commercial scale works at limited share.

WHY 50%

Modal outcome. Aurora launches commercial Driverless service on schedule (2027), proves cost-per-mile advantage on a single route (Dallas-Houston initially), expands to 2-3 routes by 2029, secures pilot partnerships with major carriers (Werner, Schneider, FedEx). Reaches ~7-8% share of $160B addressable autonomous trucking TAM by 2036 — meaningful but not dominant. Mature operating margin lands at ~20% (tech-auto blend, well below software economics). The 50% weight reflects that this is the modal outcome — most realistic for a strong technical team executing on a long technical-commercial timeline, but not requiring breakthrough commercial dominance.

WHAT HAPPENS

The Aurora Driver works at commercial scale. By FY28 the company operates 1,000–2,000 driverless trucks on commercial highways. McLane (Berkshire) expands beyond Texas. FedEx, Schneider, Werner pilots convert to multi-year volume commitments. Hirschbach's 500-truck order ships through 2027–28.

But the market splinters. Kodiak owns industrial off-highway. Waabi grows through the Volvo OEM channel. PlusAI lands legacy retrofits. Aurora becomes one of 2–3 'winners' in US for-hire long-haul, capturing ~7.5% of the $160B TAM by FY36 — $12B revenue. Mature margin settles around 20% — a blend of software-licensing and own-fleet operations.

BULL $37.32

+382.2% vs spot

Probability 12%

Dominant US platform, software-like margins.

WHY 12%

Compound bet: Driverless commercial launches on schedule AND scaled trucking fleet ramps to 5,000+ trucks by 2028 AND software-take-rate model emerges (Aurora-as-platform on partner-operated fleet) AND competitive narrowing (Kodiak underperforms, Waabi stays niche) AND autonomous trucking proves uniquely viable. Each condition independently plausible at 40-60% probability; conjunctive probability lands at 15%. The asymmetric venture-bet pattern characteristic of autonomous-platform investments — bull case pays multiples-of-current- valuation if it converges, but requires conditions to align over 5-10 years.

WHAT HAPPENS

Aurora becomes the dominant US autonomous trucking platform. FY27 driverless commercial launch proceeds on schedule, unlocking follow-on orders from major carriers (Werner, Schneider, FedEx, Hirschbach). By FY28 fleet exceeds 5,000 trucks; by FY31, 50,000+. Network effects compound — lane data, edge cases, regulatory relationships in TX/OK/NM/AZ.

The competitive set narrows. Kodiak stays viable but smaller. Waabi struggles. PlusAI runs out of SPAC capital. By FY32 only Aurora and one other player operate at-scale driverless commercial fleets. Aurora captures 22.5% of the $160B TAM — $36B FY36 revenue. Pivot to software-take-rate on OEM trucks (Volvo VNL, PACCAR). Mature margin reaches 35% — similar to Mobileye's ADAS economics.

ULTRA BULL $20.76

+168.2% vs spot

Probability 3%

Dominant US; software-take-rate.

WHY 3%

Tail of tails: commercial launch on schedule (50%) AND Aurora Driver becomes the default software on PACCAR/Volvo fleets reaching ~20% of autonomous miles / ~6–7% take-rate share of TAM by 2036 (30% conditional on launch) AND software-take-rate model wins (40%) AND multiple expansion to software/aerospace-hybrid premium (50%). Each individually plausible; joint at 3-4%. Captures the scenario where Aurora becomes a category-defining Big Tech-tier company.

WHAT HAPPENS

The ultra-bull case is that Aurora becomes the dominant US autonomous trucking platform AND wins the software-take-rate game. Commercial launch on schedule by 2027 + Aurora Driver reaches 1M+ autonomous miles by 2028 + Aurora Driver running on ~20% of US autonomous long-haul miles by 2036, monetized via per-mile take-rate (~6–7% revenue-share of TAM) with PACCAR/Volvo/FedEx fleet partners rather than capex-heavy direct ownership. By FY36 Aurora generates $8B+ revenue at 40%+ software-like margins.

Compound conditional: each piece individually plausible at 50-65%; joint at 3-4%. Reflected here at 3%. This is the scenario where Aurora joins the Tesla/Nvidia tier — not just 'autonomy works,' but 'Aurora wins the autonomy market.' Contributes $3.00 to weighted ($100 × 3%) — the asymmetric tail.

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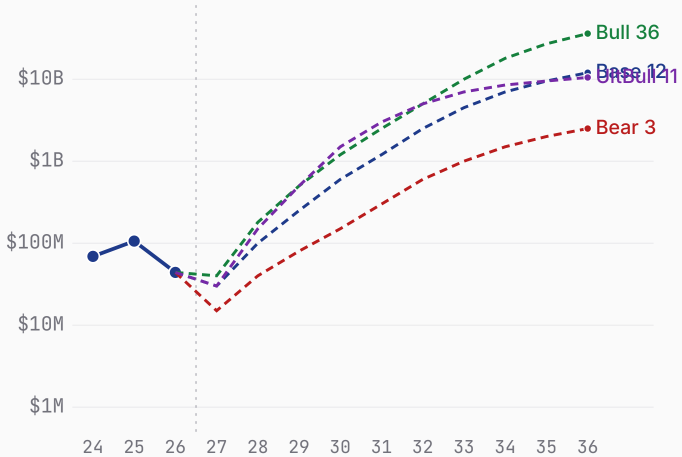
16 May 2026 · the snapshot

Aurora Innovation business snapshot

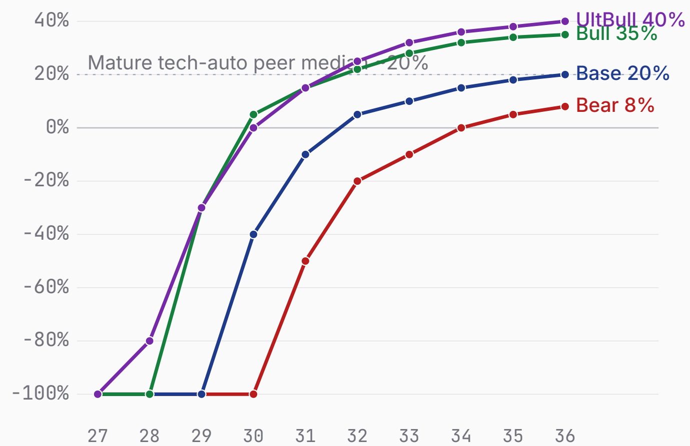
Bear $0.11 Base $3.03 Bull $37.32

FY24–FY26 history + FY27–FY36 scenario projections · fiscal years end Dec 31 · 10-K, Q1 2026 10-Q, ATA/FTR Tonnage Index

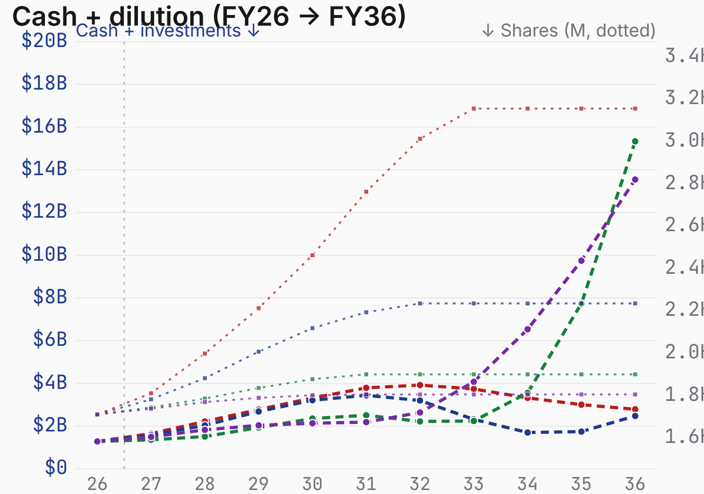
Revenue — history + scenarios to FY36 (log)



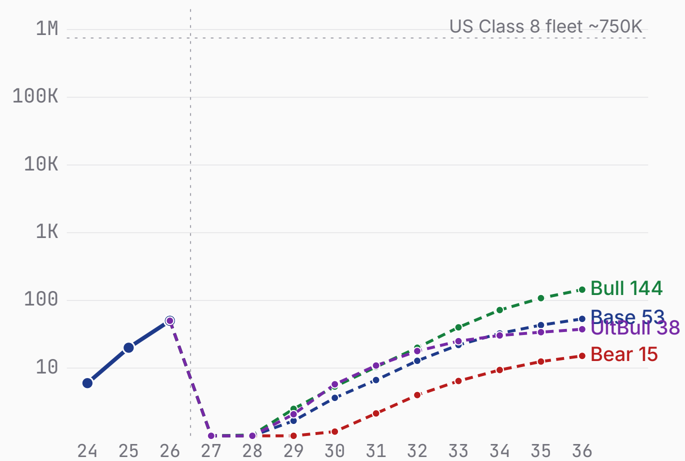
Path to profitability — op margin (FY27→FY36)



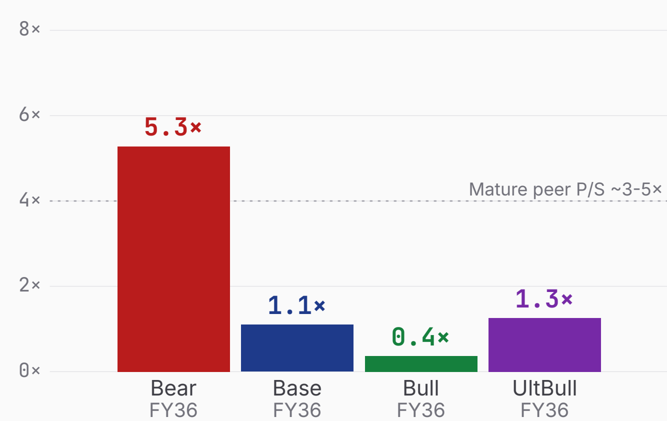
Cash + dilution (FY26 → FY36)



Driverless trucks deployed (log)

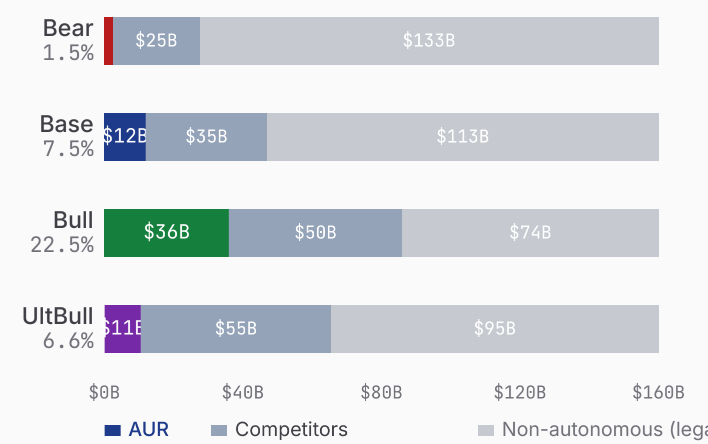


$13.20B mkt cap as P/S on FY36 rev



Bear=stretched (6×), base=fair (1.3×), bull=cheap (0.5×).

$160B US autonomous trucking TAM (FY36) — AUR share



Sources: AUR 10-K FY25, Q1 2026 10-Q, ATA/FTR Tonnage Index (US for-hire long-haul $300B; autonomy-addressable $160B).

Aurora Innovation 7 Powers · the moat, scored

Bear $0.11 Base $3.03 Bull $37.32

Arena. Driverless Class-8 trucking on US highways — Aurora vs Kodiak, Waabi, Torc/Daimler, Gatik; the race to scale commercial driverless freight.

POWER ORIGINATION
origination window: open

POWER ORIGINATION · 7 POWERS

Scale Economies	1 of 3	Per-mile cost falls with fleet, but the driverless fleet is barely launched; R&D is still a fixed sink.
Network Economies	1 of 3	Aurora Driver improves with miles driven; a data flywheel, not yet a moat.
Counter-Positioning	2 of 3	Driver-as-a-service subscription model carriers/OEMs can't easily self-build.
Switching Costs	1 of 3	Carrier/terminal integration nascent; lane-by-lane.
Branding	1 of 3	Safety-case credibility; not yet the category default.
Cornered Resource · thesis leans here	2 of 3	Aurora Driver + FirstLight lidar IP; PACCAR/Volvo/Continental platform partnerships; Dallas–Houston–Phoenix terminals.
Process Power	1 of 3	Safety Case Framework rigor; not yet a durable cost/quality edge.

THE RACE · NAMED RIVALS

Kodiak — SPAC; ~$ hundreds M
~12% terminal share · Driverless in the Permian (off-highway first); SPAC-funded.

Waabi — ~$200M raised
~10% terminal share · Generative-AI simulation approach; Volvo/Khosla-backed.

Torc Robotics (Daimler) — Daimler balance sheet
~10% terminal share · OEM-captive; Freightliner integration.

Gatik — ~$300M raised
~8% terminal share · Short-haul B2B (Walmart, Tyson); narrower problem.

LEAD / LAG · FALSIFIERS

Driverless commercial launch	live (TX) vs off-highway	LEADING
OEM platform partners	3 vs 1	LEADING
Cash runway (yrs at burn)	2 vs 3	LAGGING

COMPETITIVE READ
A genuine first-to-driverless and OEM-partnership lead, but scale and network power are unbuilt and the model is deeply capital-hungry; the window is open while Aurora out-executes — a safety or capital stumble cedes it.

Aurora Innovation show your work

Bear $0.11 Base $3.03 Bull $37.32 · Weighted $6.65 (-14.0%)

BEAR $0.11

ASSUMPTIONS

TAM Year-10 ($B)	160
Mature share (%)	1.5
Mature op margin (%)	8
Sales-to-capital	1.2
Terminal WACC (%)	10.0
Terminal growth (%)	2.0
P(failure) (%)	45
Scenario probability (%)	35

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.01	-800%	-0.13	-0.11
FY28	0.04	-400%	-0.18	-0.14
FY29	0.08	-200%	-0.19	-0.13
FY30	0.15	-100%	-0.21	-0.12
FY31	0.30	-50%	-0.28	-0.15
FY32	0.60	-20%	-0.37	-0.17
FY33	1.00	-10%	-0.43	-0.18
FY34	1.50	+0%	-0.42	-0.16
FY35	2.00	+5%	-0.32	-0.11
FY36	2.50	+8%	-0.22	-0.07

Σ PV explicit FCF	-1.35
+ PV terminal (g 2.0%)	0.72

EQUITY BUILD · $B & PER SHARE

Operating EV	$-0.63B
+ Cash & investments	$1.28B
= Equity value	$0.65B
Raised over period	$4.25B
Dilution by FY36	+74%
FY36 shares (M)	3,401
DCF per share	$0.19
× (1−P_fail) [55%]	$0.10
+ P_fail × distress [45% × $0.00]	$0.00
= Expected per share	**$0.11**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$0.18	$0.29	$0.46	$0.74
0.02×	0.04×	0.06×	0.10×

BASE $3.03

ASSUMPTIONS

TAM Year-10 ($B)	160
Mature share (%)	7.5
Mature op margin (%)	20
Sales-to-capital	1.5
Terminal WACC (%)	8.5
Terminal growth (%)	2.5
P(failure) (%)	25
Scenario probability (%)	50

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.03	-600%	-0.20	-0.18
FY28	0.10	-250%	-0.30	-0.24
FY29	0.25	-100%	-0.35	-0.25
FY30	0.60	-40%	-0.47	-0.30
FY31	1.20	-10%	-0.52	-0.30
FY32	2.50	+5%	-0.77	-0.41
FY33	4.50	+10%	-0.98	-0.47
FY34	7.00	+15%	-0.84	-0.37
FY35	9.50	+18%	-0.32	-0.13
FY36	12.00	+20%	+0.23	+0.09

Σ PV explicit FCF	-2.57
+ PV terminal (g 2.5%)	10.89

EQUITY BUILD · $B & PER SHARE

Operating EV	$8.32B
+ Cash & investments	$1.28B
= Equity value	$9.60B
Raised over period	$4.50B
Dilution by FY36	+27%
FY36 shares (M)	2,480
DCF per share	$3.87
× (1−P_fail) [75%]	$2.90
+ P_fail × distress [25% × $0.50]	$0.13
= Expected per share	**$3.03**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$4.56	$6.85	$10.30	$15.49
0.59×	0.89×	1.33×	2.00×

BULL $37.32

ASSUMPTIONS

TAM Year-10 ($B)	160
Mature share (%)	22.5
Mature op margin (%)	35
Sales-to-capital	1.8
Terminal WACC (%)	7.5
Terminal growth (%)	3.0
P(failure) (%)	12
Scenario probability (%)	12

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.04	-500%	-0.22	-0.20
FY28	0.18	-150%	-0.35	-0.28
FY29	0.50	-30%	-0.33	-0.24
FY30	1.20	+5%	-0.33	-0.22
FY31	2.50	+15%	-0.35	-0.21
FY32	5.00	+22%	-0.29	-0.16
FY33	10.00	+28%	+0.02	+0.01
FY34	18.00	+32%	+1.32	+0.60
FY35	27.00	+34%	+4.18	+1.76
FY36	36.00	+35%	+7.60	+2.97

Σ PV explicit FCF	+4.02
+ PV terminal (g 3.0%)	85.38

EQUITY BUILD · $B & PER SHARE

Operating EV	$89.40B
+ Cash & investments	$1.28B
= Equity value	$90.68B
Raised over period	$2.80B
Dilution by FY36	+10%
FY36 shares (M)	2,145
DCF per share	$42.28
× (1−P_fail) [88%]	$37.21
+ P_fail × distress [12% × $1.00]	$0.12
= Expected per share	**$37.32**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$53.58	$76.92	$110.43	$158.53
6.92×	9.94×	14.27×	20.48×

ULTRA BULL $20.76

ASSUMPTIONS

TAM Year-10 ($B)	160
Mature share (%)	6.6
Mature op margin (%)	40
Sales-to-capital	2.5
Terminal WACC (%)	8.0
Terminal growth (%)	3.5
P(failure) (%)	3
Scenario probability (%)	3

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.03	-250%	-0.10	-0.09
FY28	0.15	-80%	-0.24	-0.19
FY29	0.50	-30%	-0.50	-0.36
FY30	1.50	+0%	-0.75	-0.49
FY31	3.00	+15%	-0.75	-0.44
FY32	5.00	+25%	-0.25	-0.14
FY33	7.00	+32%	+0.84	+0.41
FY34	8.50	+36%	+1.81	+0.81
FY35	9.50	+38%	+2.61	+1.08
FY36	10.50	+40%	+3.45	+1.33

Σ PV explicit FCF	+1.93
+ PV terminal (g 3.5%)	40.40

EQUITY BUILD · $B & PER SHARE

Operating EV	$42.33B
+ Cash & investments	$1.28B
= Equity value	$42.96B
Raised over period	$2.00B
Dilution by FY36	+18%
FY36 shares (M)	2,010
DCF per share	$21.37
× (1−P_fail) [97%]	$20.73
+ P_fail × distress [3% × $0.98]	$0.03
= Expected per share	**$20.76**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$30.50	$44.82	$65.85	$96.76
3.94×	5.79×	8.51×	12.50×

Aurora Innovation supporting analysis · disclaimers · glossary

Bear $0.11 Base $3.03 Bull $37.32

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **Commercial operation is real.**
Aurora Driver runs supervised highway pilots with McLane (Berkshire), PACCAR, Volvo. Removes 'science project' bear framing.

2 **Partnership stack across OEMs.**
Volvo VNL Autonomous and PACCAR Peterbilt/Kenworth integrations. Unlike Kodiak (cargo-only) or Waabi (Volvo-only), Aurora spans both.

3 **$1.28B liquidity + capital flexibility.**
Q1 2026 raise extended runway. Strategic options (M&A interest from majors) provide downside floor.

4 **First-mover regulatory moat.**
Aurora has logged the most miles in commercial operation. Lane data, TX/OK/NM regulatory relationships compound.

5 **Mature margin may be conservative.**
Base 20% reflects tech-auto blend. If software-take-rate on OEM trucks scales (Mobileye analog), terminal could be 30%+.

FALSIFICATION TRIGGERS

Bear validation

Driverless launch slips to 2028+ · Cash < $500M by mid-2027 · Commercial volume disappoints (< 100 trucks by EOY 2028)

Bull validation

Driverless commercial launches by Q3 2027 · Fleet > 500 trucks by EOY 2028 · Second OEM software-take-rate deal signed

Reframe needed

If Kodiak achieves at-scale commercial first — Aurora's first-mover advantage in long-haul erodes; TAM-share math compresses

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Forward-looking statements.

Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.

Author may hold positions.

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Aurora Driver — Aurora's autonomy software + hardware stack. Currently runs supervised on highway pilots with PACCAR, Volvo, and FedEx.

Hub-to-hub — Highway-only routes between terminals (e.g., Dallas-Houston). Aurora's commercial focus. Easier autonomy domain than urban streets.

Driverless commercial — Operation with no human driver in the truck, generating per-mile or per-load revenue. Aurora's 2027 launch milestone.

OEM partners — Volvo VNL Autonomous and PACCAR (Peterbilt, Kenworth). OEM integration gives Aurora a manufacturing path competitors lack.

S2C (sales-to-capital) — Damodaran reinvestment ratio. ΔRevenue / S2C = reinvestment in $. Higher = more capital-efficient growth.

Take-rate model — Revenue per autonomy-mile on OEM-built trucks (vs Aurora-owned fleet). Software-like margins; the bull terminal-margin driver.